

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Requisition for Induction Bends
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-REQ-006
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Requisition for Induction Bends

R01	09.08.22	Issued for Review	I. Osobase	E.Nwachukwu	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	09.08.22	Issued For Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-REQ-006

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMG-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviations	Meaning
CIF	Cost, Insurance and Freight
CFR	Cost and Freight
EPC	Engineering, Procurement and Construction
NDT / NDE	Non-Destructive Testing / Non-Destructive Examination
ITP	Inspection and Testing Plan
PO	Purchase Order
QA/QC	Quality Assurance and Quality Control
VPIS	Vendor Print Index & Schedule
UOM	Unit of Measurement
CIF	Cost, Insurance and Freight
IOM	Installation, Operation, Maintenance Manual
RFQ	Request For Quotation

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

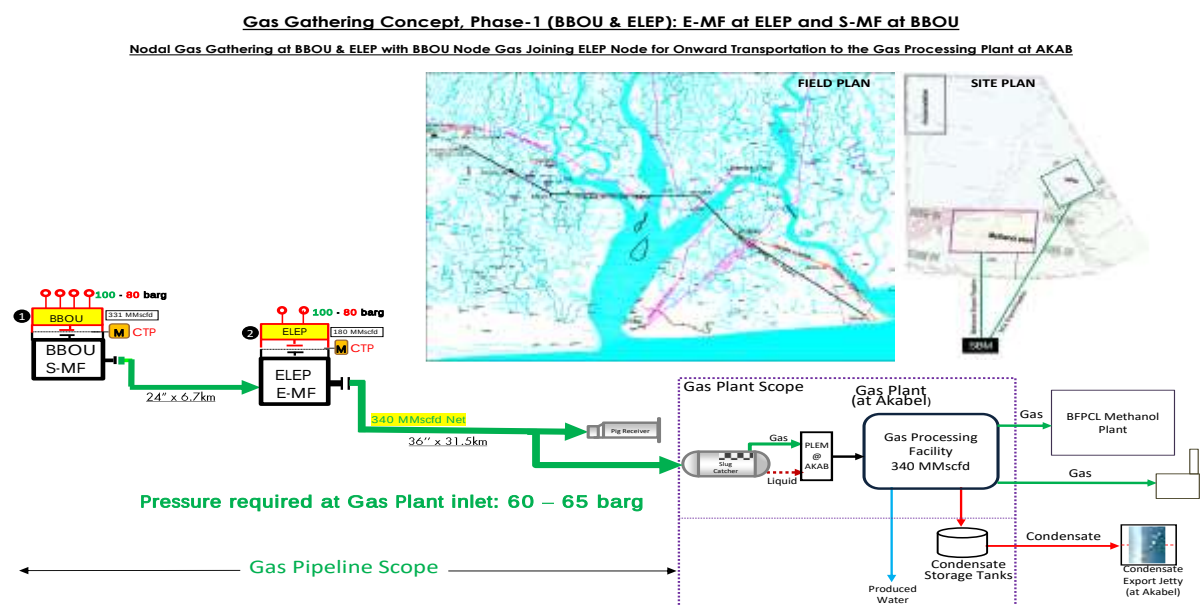


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This Material Requisition covers the scope of supply of Induction Bends coated with 3LPE coating to be used for the intersection Points (Ips) on Mainline construction and installation along the Right of Way (ROW) between BBOU Manifold station and ELEPA Manifold and between ELEPA Manifold Station and AKABEL Pigging station for the BFPCL Gas Gathering Project-Phase 1.

This document shall be read in conjunction with the Project Specifications, Reference Documents, Datasheets and related Industry Codes and Standards.

1.3. Reference Documents

S/ N	Document number	Title
1	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis for Design
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006	Specification for Induction Bends
3	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006	Data Sheet for Induction Bends
4	EA-BFPCL-GPMC-GGPL-FEED-PPL-RPT-002	Wall Thickness Calculation Report
5	DEP. 31.40.20.37-Gen	Line pipe for Critical Service (amendments/supplements to IOGP S-616 AND DNVGL-ST-F101 2019)
6	DEP. 82.00.10.10-Gen	Project Quality Assurance
7	ASME B36.10M	Welded and Seamless Wrought Steel Pipe
8	ASTM A106	Seamless Carbon Steel Pipe for High Temperature Service
9	ASME B31.8	Gas Transmission and Distribution Piping Systems
10	API 5L	Specification for Line Pipe
11	API RP 5L1	Recommended Practice for Rail Road Transportation of Line Pipe
12	API RP 5L6	Transportation of Line Pipe on Inland Waterways
13	API RP 5LW	Recommended Practice for Transportation of Line Pipe on Barges and Marine Vessel
14	BS EN 10204	Metallic Products - Types of Inspection Documents
15	ISO 3183: 2007	Petroleum and Natural Gas Industries Steel Pipe for pipeline transportation systems.

2. Definition of Terms

Terms	Definitions
Project	Gas Gathering Pipelines & Associated Infrastructure (Phase1)
Company / Purchaser	Brass Fertilizer & Petrochemical Limited (BFPCL) that initiates the project and ultimately pays for its design Procurement and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
EPC Contractor	The party who Company has employed to perform all works related to the Scope of Work of the Project.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer / Vendor / Supplier	Company or factory responsible for design and manufacture of Equipment and provides it to the Company of this Project.
Sub-Vendor	Company or factory that provides part of or complete services in the design, manufacture and supply of the equipment on behalf of the Vendor for this Project.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.

The word Manufacturer / Vendor / Supplier shall be used interchangeably in the context of this specification.

3. Line Pipe Data

3.1. 24" Multiphase Pipeline Data

The 24" pipeline design data for the Multiphase Pipeline Project are as shown below

Parameter	Description
Design Code	ASME B31.8
From	BBOU Manifold Station
To	ELEPA Manifold Station
Pipeline Length	6.7km
Design Life	30 years
Process Fluid	Condensate / Saturated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature (Vent Line)	-13.3°C
Line Pipe Material Grade	API 5L-X70(PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	24"
Density	7,850 kg/m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570 MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
Wall Thickness	15.88mm
Design Factor	0.6
Corrosion Allowance	3 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	60mm (Not in the scope of Supplier)
Installation Condition	Trenching
Cathodic Protection	Impressed current

3.2. 36" Multiphase Pipeline Design Data

The 36" pipeline design data for the Multiphase Pipeline Project are as shown below

PARAMETER	DESCRIPTION
From	ELEPA Manifold Station
To	AKABEL Station
Pipeline Length	31.5km
Design Life	30 years
Process Fluid	Condensate /hydrated Hydrocarbon Gas.
Design Pressure	110 barg
Operating Pressure	100 barg
Max Design Temperature	60°C
Min Design Temp (Pipeline)	12.7°C
Min Design Temperature	-13.3°C
Line Pipe Material Grade	API 5L-X70 (PLS2) SAWL - 100% radiographed.
Installation Temperature	23°C
Specified Minimum Yield Strength	483 Mpa
NPS	36"
Density	7,850 kg /m ³
Young's Modulus of elasticity	2.07 x 10 ⁵ MPa
Poisson's ratio	0.3
Coefficient of linear thermal expansion	11.7 x 10 ⁻⁶ m/ m/ °C
Ultimate Tensile Strength	570MPa
Steel Specific Resistivity	0.018 Ω m/mm ²
Wall Thickness	23.83mm
Design Factor	0.6
Corrosion Allowance	6 mm
ANSI Class	900#
Parent Pipe External Coating	3LPE
Concrete Coating Thickness	85mm (Not in the scope of SUPPLIER)
Installation Condition	Trenching
Cathodic Protection	Impressed current

4. Scope of Supply

- Fabrication, Inspection and Testing, surface preparation and coating with 3LPE of the Pre-fabricated bends, all necessary quality control and quality assurance incidental to the manufacture and supply of the Induction Bends coated with 3LPE to be used for Mainline installation along the Right Of Way (ROW) between BBOU Manifold station and ELEPA Manifold and between ELEPA Manifold Station and AKABEL Pigging station, producing required documentation as per the approved VPIS, handling, storage and shipping and providing total performance guarantee of the Induction Bends, in accordance with Project Specifications, Datasheets, international codes, standards and all applicable documents
- VENDOR shall provide all the equipment, labour, materials, and consumables required for the design, manufacturing, coating, inspection, testing, supply and delivery of the induction bends.

Table 1 - Scope of Supply

Item No.	Tag No.	Item Description	UOM	Quantity
1.	EQUIPMENT / MATERIAL			
1.1	Stamping of Induction Bends shall be as per Applicable Specification	Induction Bends The scope of supply shall include: Manufacture, Supply, Inspection and Testing, surface preparation for shipment, all necessary quality control and quality assurance incidental to the manufacture and supply of the INDUCTION BENDS coated with 3LPE to be used for ROW installations, producing required documentation as per the approved VPIS, handling, storage and shipping and providing total performance guarantee of the Induction Bends, in accordance with Project Specifications, Datasheets, international codes, standards and all applicable documents. The Bare Mother Pipe to be used in the fabrication of the Induction Bends shall be provided by the Company through the Line pipe Manufacture. The induction bend Vendor shall be responsible for the collection and transportation of these mother Pipes from Line pipe manufacturer's factory/warehouse to his own factory. The Induction Bends within the scope of this supply are those listed in the reference document (Induction Bend List) in Attachment 1 of this MR. For detailed scope, refer to - EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006; Specification for Induction Bends. - EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006; Data Sheet for Induction Bends.	Nos.	37
2	INSTALLATION AND COMMISSIONING SPARE PARTS			

	Not Applicable				
3	TESTING AND INSPECTION				
3.1	-	Inspection and Testing for the Induction Bends shall comply with the requirements of the under listed document and the applicable Industry Standard - EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006; Specification for Induction Bends - EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006; Data Sheet for Induction Bends.	Lot	1	
3.2	-	All Manufacturer / Supplier Material Certificates Required as per Project Specification and Industry Standard	Lot	1	
4	OTHER SERVICES (NON-MATERIAL DELIVERABLES)				
4.1	-	Engineering Services, as per requisition and Project specification.	Lot	1	
4.2	-	Submission of Documents as per VPIS in Attachment-B	Lot	1	
4.3	-	Surface preparation and protection Induction Bends for shipment shall be per Manufacturer standard	Lot	1	
4.4	-	Preservation Suitable for road transport and multiple handling. Refer to EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001; Equipment & Material Preservation Procedure. EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001; Project Quality Management Plan (QA & QC)	Lot	1	
4.5	-	Shipping Documents and Customs Clearance as per project requirements.	Lot	1	
4.6	-	Meetings requirements (Kick-off, Progress, Vendor document review and TQs etc.)	Lot	1	

4.7	-	Warranties/ guarantee as per Applicable Specification for Line Pipe and PO.	Lot	1
4.8		Estimated delivery term is CIF based on INCOTERMS 2020. (The delivery terms will be finalized before placement of PO).	Lot	1
4.9		All Induction Bends supplied shall be from Company Approved Vendor & Sub Vendor List.	Lot	1
4.10		PMI for SS and Duplex material	Lot	1
5	OPTIONAL ITEMS			
5.1		Delivery CFR Nigerian Port basis	Lot	1
6	LIST OF EXCLUSIONS			
6.1		Supply of Anchor bolts	N / A	
6.2		Installation at site / Erection	N / A	
6.3		Commissioning	N / A	
6.4		Foundation Design	N / A	

4.1. Technical Notes

All technical notes shall be read in conjunction with the following documents

- EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006; Specification for Induction Bends
- EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006; Data Sheet for Induction Bends.

5. Scope of Work

5.1. General

Supplier's scope of work includes Manufacture, Supply, Inspection and Testing, qualifications, Coating with 3LPE, preparation and preservation for seaworthy transportation, producing required documentation, handling, storing and shipping and providing total performance guarantee of the Induction Bends, in accordance with data sheets, relevant Specifications, international codes, standards and all applicable documents.

5.2. Technical Coordination Meeting

After the order is awarded, Supplier shall, upon receipt of purchaser's notice, send a sufficient number of qualified personnel to Purchaser's office at Supplier's cost to hold the kick-off meeting not later than 2 week of purchase order and the design clarification meetings.

5.3. Delivery Schedule

Suppliers to submit the best delivery schedule for the Induction Bends. Delivery schedule offered by Suppliers will be one of the justification factors while Technical and Commercial Bid Analysis is performed by Company for the purposes of determining successful Supplier.

5.4. Project management

VENDOR shall constitute a dedicated Project Team which shall coordinates all the activities and services throughout the Manufacturing and Supply cycle including establishment of overall Project Execution plan compliance with contractual, statutory & regulatory requirements and Technical/commercial logistic requirements.

5.5. Engineering Services

The final design of Induction Bends shall be the sole responsibility of the Manufacturer/Supplier with the regard to bend angles listed in attachment 1. The Supplier shall ensure that the final design of the Induction Bends complies fully with the design criteria as per project requirement and applicable specifications and data sheets. The Supplier shall also be responsible for certification and design of all necessary accessories required for the transportation of the Induction Bends.

The VENDOR is responsible for the fabrication, inspection/testing, supply, guarantee and documentation of all the Induction Bends. This shall include, but not limited to:

- a) Relevant Drawings and Documentation.
- b) Resolving engineering technical queries relating to Induction Bends within the scope of supply.
- c) Expediting services for the sub-contracted items. Ensuring that Sub-VENDORS shall adhere to project requirements specifications and datasheet.
- d) Participating in co-ordination meetings and kick-off meeting.

5.6. Design Changes

Any design changes recommended by SUPPLIER, which shall affect the Purchase Order and/or Specifications, drawings, etc. shall be brought to CONTRACTOR through COMPANY's written approval before proceeding with the work.

5.7. Supply of spares as per Project Scope.

Nat Applicable

6. Information Required along with Bid

VENDOR shall submit all the details including applicable Specification for the Induction Bends, Datasheet and Drawings completed with vendor offer, and other project documents in their technical offer.

Prospective VENDORS shall ensure that they do not undervalue the ratings, quality or quantity stated in the RFQ to make their offer competitive offers.

VENDOR shall include in the technical bid, his previous experience in the manufacture and procurement and supply of similar Induction Bends.

6.1. Deviations and Alternatives

- The deviation list shall be prepared by VENDOR and shall be included in Vendor's bid/proposal (stamped and signed scan copy). in any case where VENDOR has no exceptions, VENDOR shall stamp and sign clearly stating "No Deviations" in VENDOR deviation form, technical explanations for proposed deviations shall be included in VENDOR deviation form as per Attachment D.
- VENDOR shall be responsible for any deviation or alternatives included in VENDOR'S bid/proposal and not stated in the deviation list. Any such hidden deviations shall not be entertained. Vendor is requested to submit filled in "Deviation List" and a completed technical checklist in the format attached in Attachment D and E.
- Assumptions, Alternatives, Deviations, Exceptions and Concessions raised shall be recorded and supplemented with adequate technical justifications. If no deviation list in the required format is furnished with the Bid, then it shall be construed that Vendor has no deviations to the requisition and its attachments and the Bid is in full compliance.
- After placement of Purchase Order, no deviation shall be acceptable from VENDOR.
- Vendor to provide the filled-out form as per Attachment C and Completed technical checklist along with bid.

7. Supplier proposal submittal

7.1. Supplier Proposal

The Supplier shall prepare the proposal in accordance with the requirements set forth in the RFQ documents. Any proposal not conforming to the RFQ and attached documents may be disqualified. However, Supplier may submit his proposed deviations for consideration by Company.

7.2. Number of Copies

The Supplier shall submit 1 (one) original and 2 (two) copies of the proposal, consist of:

- a) Section 1: Cover Letter Stating the following:
 - SUPPLIER'S agreement with the Terms and Condition in the associated RFQ
 - Total value of quotation price (excluding VAT)
 - SUPPLIER'S statement that the work can be started immediately on issuance of Letter of Intent.
- b) Section 2: Schedule of Prices
 - A Supplier shall submit detail of price breakdown and also summary of unit rates as required.
- c) Section 3: Detail Schedule
 - SUPPLIER shall submit a detailed schedule for the work and/or delivery required to supply the quantity of data requested.
- d) Section 4: Quality Control Requirement
 - SUPPLIER' Inspection and Test plan procedures to assure that quality of the materials meets the specifications
- e) Section 5: Others (minimum as per the Data Sheet, Material Requisition for Induction Bends)
 - In this section, SUPPLIER shall submit other documents requested by Company and other supporting documents for this requisition.

7.3. CLOSING DATE

Supplier shall submit the proposal to Company within 2 (two) weeks after receiving the RFQ documents

7.4. SCHEDULE

Contractor requires delivery completed in 4 months. Suppliers to submit its best delivery schedule for the supply of the Induction Bends.

ATTACHMENT A - Technical Documents and Attachments

Table 2 - Technical Documents

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006	Specification for Induction Bends
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006	Data Sheet for Induction Bends
EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001	Equipment & Material Preservation Proce-
EA-BFPCL-GPMC-GGPL-FEED-SCM-PLN-003	Procurement & Subcontracting Plan.
EA-BFPCL-GPMC-GGPL-FEED-HSE-PLN-001	Project HSE Plan
EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001	Project Quality Management Plan (QA & QC
ATTACHMENT 1	MTOs for Induction Bends

ATTACHMENT B - Vendor Drawing and Data Requirement List

Table 3 - Vendor Drawings and Data Requirement List

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Re-quired for Final submission	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
1.	Vendor Print Index & Schedule	Y	Y	2 Weeks	Y		3C + 3E Flash-Driver / HardDisk	
2.	Vendor / Manufacturer's Catalogue	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / HardDisk	
3	Project baseline detailed schedule.	Y	Y	4 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
4	Supplier experience list	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
5	progress Measurement reports / charts			Weekly	Y		3C + 3E Flash Driver / Hard Disk	
6	Approved sub-vendor list (and places of manufacturing/ inspection)	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
7.	Project Quality Plan, QAC Procedures	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
8	Sub order schedule		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
9	Preservation, packing, handling and shipping procedure	Y	Y	4 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
10	Detail fabrication drawings for main items		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Re-quired for Final submission	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
11	Destructive and non-destructive test procedures including NDT, Operators list (Certification level and validity date)		Y	3 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
12	Manufacturing Procedure Specification (MPS) Manufacturing procedure qualification test (MPQT) results, manufacturing procedure qualification test (MPQT) results		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
13	Inspection and Test Plan	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
14	Certified Welding Procedure Specification (WPS), Procedure Qualifications (WPQ) and Welders Performance Qualifications Record (WPQ)		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
15	Certified Hydrostatic Test Report		Y	1 week before inspection release note	Y		3C + 3E Flash Driver / Hard Disk	
16.	Material Test Reports		Y	1 week before inspection release note	Y		3C + 3E Flash Driver / Hard Disk	
17.	NDT/ NDE Test Reports		Y	1 week before inspection release note	Y		3C + 3E Flash Driver / Hard Disk	

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Re- quired for Final submis- sion	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
18	Deviations	Y		With Bid			3C + 3E Flash Driver / Hard Disk	
19.	Sub order copies (un-priced)			On regular basis	Y		3C + 3E Flash Driver / Hard Disk	
20.	PWHT Procedure, as applicable		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
21.	Certification Data Book		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
22.	Material test certificate as per EN 10204 as mentioned in Data Sheet		Y	4 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
23	Shipping Documents and Transport Authorization (Final Packing List + Packing Content + Commercial Invoice + Certificate of Origin + Phytosanitary Certificate)		Y	6 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
24	Vendors Document control Manual		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
38	Project QA/QC record		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
25	Spare parts List for Erection, Pre-commissioning and Commissioning		N/A					

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Re-quired for Final submission	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
26	Two (2) Year Operating Spare Parts List						N / A	
27	Special Tools for Erection, Pre-Commissioning, Commissioning and Operation List						N / A	
* After P.O. ** From P.O. Date Legends: C: Copy E: Electronic Copy Note: 1. The above-mentioned documents are general requirements; the same shall be discussed and agreed with Company / its representative and Vendor, prior to Placement of Order. 2. Electronic copy required at all stages. 3. One copy of IOM document shall be submitted along with Package delivery.								

ATTACHMENT C - Table of Compliance

VENDOR shall confirm below the Sections of the Requisition/Specification that he has complied with/accepted or further clarifications are desired, or VENDOR has taken a deviation with reasons thereto. Columns 2, 3, and 4 must be answered and initialed.

Table 4 - Table of Compliance

1	2	3	4
Specification/Datasheet No.	*Conforms Yes/ No	*Deviations Yes/ No	@Clarification or Reasons for Deviations
Specification for Induction Bends: EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-006			
Data Sheet for Induction Bends: EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-006			
Equipment & Material Preservation Procedure: EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001			
Procurement & Subcontracting Plan: EA-BFPCL-GPMC-GGPL-FEED-SCM-PLN-003			
Project HSE Plan: EA-BFPCL-GPMC-GGPL-FEED-HSE-PLN-001			
Project Quality Management Plan (QA & QC): EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001			

Notes:

- Acceptance (i.e. yes) implies total compliance to the Specifications.
- Provide additional explanatory sheets if required.
- Above are the main documents to be complied. However, compliance is required on all documents referenced in this requisition.

ATTACHMENT D - Deviation List

Table 5 - Deviation List

Sr. No	Docu-ment	Refer-ence / Par-agraph	Devia-tion / Clari-fication	Justification for Devia-tion / Clarifi-cation	EPC Con-tractor Ap-proval	COM-PANY Ap-proval	Remarks
1							
2							
3							
4							
5							
6							

Notes:

In case of no deviations, VENDOR shall state "NO DEVIATIONS" with stamp and sign below.

- Deviations listed elsewhere in VENDOR's offer shall not be considered. VENDOR shall be responsible to comply at his own cost and time.
- Offers that do not include signed and stamped Deviation List shall be liable for rejection.
- No deviations from any part of this requisition are allowed unless stated in VENDOR's bid using the above format.

ATTACHMENT E -Technical Checklist

VENDOR shall ensure that the following minimum information is attached with their bid.

Table 6 - Technical Checklist

S. NO	DESCRIPTION	PLEASE REPLY YES/ NO
1.	VENDOR to furnish Quotation Reference and Quotation	
2.	Completely filled in 'Table of Compliance'.	
3.	Deviations, if any, to the scope of work, specification, codes and standards attached / referred to in this requisition with proper reasoning.	
4.	Completely filled in data sheets.	
5.	Signed and Stamped Specifications	
6.	VENDOR's quality control documents (viz. Manufacturing, inspection and testing)	
7.	Compliance that delivery schedules and VENDOR's detailed order execution schedule is submitted.	
8.	QA/QC documents as required with bid.	
9.	List of Special Tools and Tackles (if applicable)	
10.	Supplier to Guarantee the performance parameters as per the specifications and datasheets.	
11.	Supplier to confirm that quality certificate of the manufacturer facility and quality plans/ inspection and test plans is furnished with the bid.	
12.	Any additional documents/ data as required by Supplier.	
13.	Supplier to quote for all items (type & design etc.) as per material requisition scope of supply and duly filled in signed and stamped and attached with the Offer.	
14.	Supplier to confirm that the offer is in full compliance with all requirements of Inquiry notes, Specifications / Standards attached with Inquiry, MTO for Induction Bends as per Attachment-1	
15.	Supplier to confirm that Unit price, total price and delivery period has been filled in, in the Material Requisition Scope of Supply.	

16.	Supplier to confirm that quotation shall be applicable and valid for placement of order either in full or in part based on the unit rates quoted.	
17.	Supplier to confirm that shipping weights and volume has been mentioned in the offer.	
18.	Supplier to confirm that they are aware of all arrangements for testing and inspection, in compliance with the Inquiry/specification, at Supplier's work shall be done by them.	
19.	Supplier to indicate Spare part(s) for Two years continuous Operation, Commissioning spare(s) & Maintenance spare(s) (Itemized price list) if any	Not Applicable
20.	Supplier to confirm that they have indicated the country of origin and is aware that the certificate of origin shall be part of shipping documents.	
21.	Supplier shall indicate the Production / Delivery schedule.	
22.	Supplier to confirm that they have attached technical catalogue or product brochure with the offer.	
23.	Supplier to attach their "Quality Management System" along with the quotation.	
24.	Supplier to confirm that the typical ITP (Inspection and Test Plan) for the quoted items has been attached along with the quotation.	
25.	Supplier to confirm that Certificates for ISO accreditation for Self and Sub Suppliers are attached with offer	
26.	Supplier to furnish the reference list for the past jobs executed with other customers along with customer name, project name, Items supplied qty of orders etc.	
27.	Supplier to state the warranty period and confirm that the materials shall be replaced at free of cost in case any manufacture / transit defects found during that period.	
28.	Supplier to confirm that the components quoted by them are suitable for the life span of 30 years.	
29.	Manufacturing / Production Schedule.	
30.	Tagging / Stamping of Induction Bends shall be as per as provided by Company	
31.	VENDOR to indicate Country of Origin.	

ATTACHMENT F - Inspection and Test Requirements

RFQ No:	PO No:	VENDOR:
Description and TAG No. REQUISITION FOR INDUCTION BENDS		

NOTE: FINAL CERTIFIED DRAWINGS AND DOCUMENTS MUST BE SUBMITTED WITHIN TWO WEEKS OF MANUFACTURING / FABRICATION AND TESTING COMPLETION

Table 7 - Inspection and Test Requirements

SCOPE: INDUCTION BENDS				Test and Inspection Requirements P. O. Specs and Respective Codes / Standards	
1 Visual Inspection Witnessing by EPC Contractor and Company (Note-1)					
2 Certificates / Records to be Checked by EPC Contractor & Company (Note-1)					
3 Certificates / Data to Be Provided by VENDOR / Sub-Contractor					
01		X	X	01	Manufacturing procedure specification (MPS), Qualification Tests, as applicable
02		X	X	02	Material Test Reports, Certificate of Compliance for the Equipment
03	X	X	X	03	Positive Material testing, as applicable
04		X	X	04	Heat Treatment (Certificates), as Applicable
05		X	X	05	Heat Treatment (Charts), as Applicable
06		X	X	06	Welders Qualification Records, as Applicable
07		X	X	07	Welding Procedure Specifications, Procedure Qualification Records and Weldability Test, as Applicable
08	X	X	X	08	Workmanship, Material, Surface Defects, Components, Dimensions
09		X	X	09	Compliance with Purchase Order Description, Specifications, Certified Drawings, VENDORS Catalogue Description
10		X	X	10	Quality Control Manual - Check for System Implementation
11		X	X	11	Painting /Coating Inspection, as Applicable
					Non-Destructive Examinations (NDE)
12		X	X	12	Radiographic Testing of Welds, as Applicable
13		X	X	13	Magnetic Particle Examination of Bevel Ends, as Applicable
14		X	X	14	Ultrasonic Examination, as Applicable
15		X	X	15	Others as per Specifications

SCOPE: INDUCTION BENDS					Test and Inspection Requirements P. O. Specs and Respective Codes / Standards
1 Visual Inspection Witnessing by EPC Contractor and Company (Note-1)					
2 Certificates / Records to be Checked by EPC Contractor & Company (Note-1)					
3 Certificates / Data to Be Provided by VENDOR / Sub-Contractor					
					Tests (As approved procedure)
16	X	X	X	16	Dimensional & Visual checks on all Induction Bends
17		X	X	17	100 % Calibration / Configuration checking
18	X	X	X	18	100 % Hydrostatic Testing
19		X	X	19	Testing of all other materials as per specification and all associated reference documents in this MR for more details
20	X	X	X	20	All key items and inspections and Tests as per the Project Specification & related Codes and Standards: • Testing of all Induction Bends • Mechanical Testing • Functional test operation • Painting /Protective Coating Inspection, etc.
					Other Requirements
21	X	X	X	21	Marking, Packing
22		X	X	22	Weighing
23		X	X	23	Compliance with Project Specification Requirements
24	X	X	X	24	Packing and Protection
25		X	X	25	Final Inspection and Issue of Release Note and Inspection Certification
26		X	X	26	Operating Manual, MRB, Installation and Maintenance manual
Inspection Level Required					I: II: X III: IV:
Notes: 1. VENDOR shall notify Company with min. (2) weeks in advance to witness the testing. 2. VENDOR to check the quality control requirement and Inspection Level in this table based on his experience, and further to advise, if there are deviations from VENDOR, VENDOR must seek Company approval and provide the relevant reason. Legend: X- Required					

ATTACHMENT 1 - Induction Bends List

Attachments	Document Title
ATTACHMENT 1.	Induction Bends List



ATTACHMENT%201-
%20INDUCTION%20